WASTEWISE-ML · REGULATORY AND STANDARDS REGISTER

WasteWise-ml regulatory and standards register

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DOCUMENT CONTROL

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Author	Amish Chadha	License	MIT
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/WasteWise-ml		
Source	docs/07-regulatory.md @ 9031e25		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-25	Amish Chadha	First regulatory and standards register (kit 1.8 upgrade), for a first pilot with a waste picker cooperative in Pune, India

WasteWise-ml regulatory and standards register

Which standards and regulations apply, where, and what is needed, for a first pilot in Pune, India (first candidate partner SWaCH, decided 2026-10-02). WasteWise-ml is software and a sorting station built from bought items and a small made stand; it advises and never certifies material, so most rules apply to the site and the partner rather than to the product. Not legal advice; anything marked "to assess" needs a qualified reviewer before any product claim. Rules marked "not checked in this session" are named from general knowledge and are to be read and cited before they are relied on. Method: STANDARDS section 23.

Table 1. Regulatory and standards register.

#	Standard or regulation	Jurisdiction	What it covers	Applicability to this design	Owner (role)	Status
G1	Solid Waste Management Rules, 2026 (supersede the 2016 rules; in full effect from 2026-04-01)	India	Segregation at source into wet, dry, sanitary and special care waste; dry recyclables to material recovery facilities	The station sorts dry recyclables; batteries, paint cans and similar items are special care waste that goes to designated collection points, which matches the hazard route	Partner (site operator)	Applies (site); to assess how the rules recognise and integrate waste pickers
G2	Digital Personal Data Protection Act, 2023, and Digital Personal Data Protection Rules, 2025 (notified 2025-11-13; notice and consent duties apply 18 months after notification, from about 2027-05-12)	India	Notice, consent and duties of data fiduciaries for digital personal data	Field photos are taken with consent, keep no faces and are uploaded only on opt-in (D6, PR-014); a photo of a hand or a voice prompt record could still be personal data	Data lead with the partner	To assess (met by design in principle; protocol to be written with the partner and reviewed)
G3	Legal Metrology Act, 2009, and state enforcement rules	India	Weights and measures used in any transaction must be verified and stamped; electronic and platform scales are re-verified every year	The platform scale (BOM line 7) weighs sorted lots for sale	Partner (site operator)	Applies: buy a verified and stamped scale and re-verify it yearly (BOM line 7 specification)
G4	Battery Waste Management Rules, 2022	India	Collection and handling of waste batteries	Cells and battery-powered items go to the hazard box, then to a collection route the partner and municipality agree	Partner; municipality	To assess (not checked in this session)
G5	E-Waste (Management) Rules, 2022	India	Electrical and electronic waste	Devices with batteries found in the stream; end-of-life phones and power banks of the station	Partner	To assess (not checked in this session)
G6		India				

#	Standard or regulation	Jurisdiction	What it covers	Applicability to this design	Owner (role)	Status
	Bio-Medical Waste Management Rules, 2016		Sharps and medical waste	Syringes and other medical waste found in the stream go to the sharps container; the app flags them, never sorts them	Partner; municipality	To assess (not checked in this session)
G7	Plastic Waste Management Rules, 2016, as amended (extended producer responsibility)	India	Producers' obligations and recycler registration	Context for the buyers of sorted plastics; no obligation on the app	Partner; buyers	Does not apply to the design; to assess as market context (not checked in this session)
G8	BIS Compulsory Registration Scheme for electronics	India	Safety registration of listed electronic products sold in India	The bought phone, power bank and shutter button	Procurement (partner)	To assess: buy registered products where the category is listed (scheme page https://www.bis.gov.in/product-certification/products-under-compulsory-certification/scheme-ii-registration-scheme, not read in this session)
G9	Radio equipment approval for the Bluetooth shutter button	India	Approval of short-range radio devices	BOM line 12	Procurement	To assess (not checked in this session)
G10	Dataset licenses: TrashNet (MIT), TACO (annotations CC BY 4.0, images per image), ZeroWaste (CC BY-NC 4.0)	International	Use of third-party images for training and publishing weights	ZeroWaste excluded from training (N2, decided); TACO filtered image by image (PR-013)	Data lead	Applies; at risk until the TACO image check is done
G11	Model cards for model reporting (Mitchell et al. 2019, practice)	International	Published per-class results, intended use and known failure modes	Required by PR-013	ML lead	Met by design (planned with the first released weights)
G12	ISO 11226:2000 (static working postures) and ISO 9241-303:2011 (display legibility)	International	Upper arm elevation screen; minimum	Used as screens in WML-CAL-001 section 12 (fix F1)	Design lead	Met on paper (standards not purchased; values to confirm against the texts)

#	Standard or regulation	Jurisdiction	What it covers	Applicability to this design	Owner (role)	Status
			character height			
G13	Resin identification codes (ASTM D7611)	International	The numbered marks on plastic items	Read by the camera where legible; ground truth for PR-002	ML lead	Applies (reference; cited through WasteWise Scan)
G14	EU Artificial Intelligence Act	European Union	Obligations by risk class of an AI system	Only if the app is deployed in the EU; not for the Pune pilot	Project lead	Does not apply to the pilot; risk class to assess before any EU use

Status: to assess, applies, does not apply, met by design, certification needed later.

Sources

- AZB and Partners. "India's Digital Personal Data Protection Act: Phased Rollout and Key Compliance Milestones." 2025-11-14 (DPDP Rules notified 2025-11-13; consent obligations after 18 months, about 2027-05-12). https://www.azbpartners.com/bank/indias-digital-personal-data-protection-act-phased-rollout-and-key-compliance-milestones/. Checked 2026-10-25.
- Department of Weights and Measures, Government of NCT of Delhi. Frequently asked questions (verification and stamping; re-verification intervals). https://weightnmeasures.delhi.gov.in/faqs. Checked 2026-10-25. Legal Metrology Act, 2009: https://www.indiacode.nic.in/bitstream/123456789/6927/1/legal_metrology_act_2009.pdf (not read in this session).
- Insights IAS. "Solid Waste Management (SWM) Rules, 2026." 2026-01-29 (from PIB; in full effect from 2026-04-01; four segregation streams). https://www.insightsonindia.com/2026/01/29/solid-waste-management-swm-rules-2026/. Checked 2026-10-25.
- Mitchell, M., et al. 2019. "Model Cards for Model Reporting." *Proceedings of FAT* 2019*. arXiv:1810.03993.
- Dataset licenses as recorded in `ml/data/SOURCES.md` and WML-CAL-001 section 16 (checked 2026-09-25).
- ASTM D7611 as cited in WasteWise Scan WSC-PRB-001.